Akshara Pandey

Lucknow | Gmail | +91 9839960816 | Portfolio | Linkedin | Github

Education

Dr. APJ Abdul Kalam Technical University, B.Tech in CS **CGPA: 8.29/10** Sept 2022 – May 2026
BSS Public School: Class XII: 93.8% | Class X: 95.4%
Coursework: Data Structures, Operating Systems, DBMS, OOPs, Discrete Math.

Technologies

Languages: Python, Java, C, C++, SQL (PostgreSQL, MySQL), Typescript, JavaScript.

Backend and Systems: Node.js, NestJS, Express js.

Frontend and Design: React.js, Tailwind CSS, HTML/CSS.

Tools and Devops: Git/GitHub, Jira, Docker, AWS, Postman.

Databases: MySQL, MongoDB.

Technical Experience & Case Studies

Software Engineer Intern, Tezbytes March 2026
- Built real-time notification center improving user engagement for 500+ daily active field managers through instant order status updates and read/unread tracking
- Developed backend CSV streaming export handling 1000+ order records, enabling offline analysis and reporting for field operations teams
- Architected order execution dashboard handling 100k+ records, reducing query response time from 10s to <2s through server-side pagination, database indexing on high-traffic columns, and Redis caching layer
- Implemented order lifecycle validation enforcing state machine logic (Pending → Processing → Completed), preventing invalid transitions and maintaining data integrity across multi-tenant architecture
- Optimized dashboard API responses by adding pagination metadata (total counts, page info), reducing frontend computation by 40% and improving load performance.
- **Technologies:** NestJS, React, Prisma ORM, Supabase, PostgreSQL, TypeScript, REST APIs

Data Quality Validation Project Jan 2026
- Built Python pipeline to audit 5000+ media assets in ASR dataset, identifying 118% metadata discrepancy and preventing corrupted data from entering model training

Hackathons & Open Source

Amazon Nova AI Hackathon | Nova-Act: AI Government Service Agent **March 2026**
- Built AI agent using Amazon Nova Lite and AWS Bedrock to automate government service interactions (DMV, Tax, Benefits), reducing average session time from 47 minutes to <60 seconds through Planner-Executor-Verifier workflow
- Developed real-time agent visualization dashboard in Next.js 14 showing autonomous API calls and tool execution
- **Tools Used:** AWS Bedrock, Amazon Nova, Next.js 14, REST APIs

Projects

MultiThreaded Proxy Web Server Github Link
- Built a multithreaded proxy server in C to handle concurrent client requests using POSIX threads and synchronization mechanisms.
- Managed shared resources using mutex locks and synchronization primitives to prevent race conditions and ensure thread safety.
- **Tools Used**: C, Socket Programming, Multi-threading, Synchronization primitives.

AlgoPanel | AI Mock Interview Platform Github Link
- Built full-stack AI interview platform with real-time voice interactions, resume analysis, and ATS scoring using Google Gemini 2.0 and Vapi AI for human-like conversational practice
- Implemented WebRTC-based voice interview system generating role-specific questions dynamically based on target position, tech stack, and seniority level with live transcription
- Developed resume intelligence engine parsing PDF/DOCX files, identifying skill gaps, calculating ATS compatibility scores, and recommending personalized learning resources
- Engineered interview recording infrastructure with webcam capture, video storage, and performance tracking dashboard for self-review and progress analytics
- **Tools Used:** Next.js 15, React 19, Firebase, Google Gemini 2.0, Vapi AI, WebRTC, TypeScript

Certifications

Hackerrank Software Engineering Certification Test Certification Link